Frequency-Response Models

Frequency-response models obtained using spectral analysis

Apps

System Identification	Identify models of dynamic systems from measured data

Functions

`etfe`	Estimate empirical transfer functions and periodograms
`spa`	Estimate frequency response with fixed frequency resolution using spectral analysis
`spafdr`	Estimate frequency response and spectrum using spectral analysis with frequency-dependent resolution
`idfrd`	Frequency-response data or model
`getpvec`	Model parameters and associated uncertainty data
`setpvec`	Modify value of model parameters
`bode`	Bode plot of frequency response, magnitude and phase of frequency response
`bodemag`	Bode magnitude response of LTI models
`freqresp`	Frequency response over grid
`chgFreqUnit`	Change frequency units of frequency-response data model

Examples and How To

Estimate Frequency-Response Models in the App

You must have already imported your data into the app and performed any necessary preprocessing operations.

Estimate Frequency-Response Models at the Command Line

You can use the `etfe`, `spa`, and `spafdr` commands to estimate spectral models.

Concepts

What is a Frequency-Response Model?

A *frequency-response model* is the frequency response of a linear system evaluated over a range of frequency values.

Data Supported by Frequency-Response Models

Characteristics of data supported for estimation of spectral analysis models.

Selecting the Method for Computing Spectral Models

How to select the method for computing spectral models during estimation in the app and at the command line.

Controlling Frequency Resolution of Spectral Models

Controlling frequency resolution of spectral models during estimation in the app and at the command line.

Spectrum Normalization

The *spectrum* of a signal is the square of the Fourier transform of the signal.